

```

1  %-----
2  % Regression
3  %-----
4
5  clear
6  clc
7
8  %Printing the outputs as a text file if ==1
9  output_file=0;
10
11 %Sample size
12 n=500;
13
14 %True parameter values
15 beta_1=1;
16 beta_2=3;
17 beta_vec=[beta_1 beta_2]';  %(2,1)-vector
18
19 %Generating observations
20 rng(0);          %Random number generator (MATLAB)
21 x=randn(n,1);    %(n,1)-vector of iid standard normal independent variables
22 e=randn(n,1);    %(n,1)-vector of iid standard normal error terms
23 X=[ones(n,1) x]; %(n,2)-matrix
24 y=X*beta_vec+e;  %(n,1)-vector
25
26 %Estimating best beta
27 beta_hat=inv(X'*X)*X'*y; %or: beta_hat=(X'*X)\(X'*y); Projection
28 beta_hat_2=regress(y,X); %MATLAB function regress
29
30 %Creating the regression line for Figure
31 x_min=-4;
32 x_max=4;
33 x_ax=linspace(x_min,x_max,100);
34
35 if output_file==1
36     dfile ='regression.txt';
37     if exist(dfile, 'file')
38         delete(dfile);
39     end
40     diary("regression.txt")
41     diary on
42 end
43
44 %Output (Command Window)
45 fprintf('Projection: beta=(%f,%f)\n',beta_hat(1),beta_hat(2));
46 fprintf('MATLAB Function regress: beta=(%f,%f)\n',beta_hat(1),beta_hat(2));
47
48 if output_file==1
49     diary off
50 end
51
52 %Figure
53 figure()
54 hold on
55 plot(x,y,'. '); %or: scatter(x,y);
56 plot(x_ax,beta_hat(1)+beta_hat(2)*x_ax);
57 hold off
58 xlabel('$x$', 'Interpreter', 'latex', 'FontSize', 12);

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59 ylabel('$y$', 'Interpreter', 'latex', 'FontSize', 12);
60 xlim([x_min x_max]);
61 ylim([beta_1+beta_2*x_min beta_1+beta_2*x_max]);
62 title('Regression: Example', 'Interpreter', 'latex', 'FontSize', 14);
63 %Putting the regression equation into the figure
64 reg_eq=sprintf('$y$ = %.3f + %.3f $x$', beta_hat(1), beta_hat(2));
65 text(-3.75, 2, reg_eq, 'Interpreter', 'latex', 'FontSize', 12);

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